Optimal Allocation of Observations across Run-In, Treatment, and Common Close Phases in Longitudinal Clinical Trials

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Abstract

Background. Trial designers planning a longitudinal study with a fixed total observation budget face an allocation question: how should observations be distributed across run-in (J_0), treatment (J_1), and common close (J_2) phases to minimise the variance of the treatment-effect estimator? Existing guidance is heuristic; the analytical optimum has not been previously characterised.

Methods. We formulate the allocation question as a discrete optimisation problem over the Woodbury-based variance formula derived in Paper 1 of this compendium. The solution is characterised as a function of the signal-to-noise ratio σ_b/σ. Extensions cover three settings: non-equally-spaced designs in which the timing of observations within each phase is also optimised; replicated endpoint observations (closely spaced measurements at the end of treatment that average out measurement error on the final assessment, analogous to run-in averaging applied at the post-treatment end); and pattern-mixture handling of dropout following the Frost-Kenward-Fox (2008) framework.

Results. The optimal allocation depends primarily on σ_b/σ. The common close phase is beneficial only when this ratio exceeds a threshold that depends on J_1; below the threshold, all observations should go to the treatment phase. Non-equally-spaced designs improve efficiency by 5-15% over equally-spaced designs at matched J, with the largest gains at low signal-to-noise ratios. Replicated endpoint observations reduce measurement-error variance proportionally to their count, with diminishing returns above 3-4 replicates.

Conclusions. The optimal-allocation rules derived here give trial designers a closed-form decision procedure driven by the signal-to-noise ratio. The procedure generalises to non-equally-spaced and dropout-adjusted designs without sacrificing closed-form tractability.

Contents

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1 Introduction

[1] **Additional run-in and common close observations both improve efficiency but with diminishing or J_1-dependent returns.**

- The first run-in observation yields the largest variance reduction.

- Subsequent run-in observations contribute progressively less.
- Common close gains depend on the number of treatment observations J_1.

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The relative efficiency surface in Paper 1 (Figure 1) demonstrates diminishing returns from additional run-in observations: the first run-in observation provides the largest variance reduction, while subsequent observations contribute progressively less. Similarly, common close observations improve efficiency but at a rate that depends on the number of treatment observations J_1.

[2] **The efficiency results motivate an allocation question left open by Paper 1.**

- Trials operate under constraints on total visits or total duration.
- Paper 1 characterises efficiency but not the optimal phase split.
- The question is how to distribute observations across the three phases.

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These observations raise a design question that Paper 1 does not address: given operational constraints on the total number of visits or total trial duration, what is the optimal distribution of observations across the three phases?

[3] **Prior work addresses related design questions but not optimal allocation across run-in, treatment, and common close phases.**

- Existing optima target time-point positioning or general design guidelines, not the run-in setting.
- Multiple pre-treatment measurements are known to benefit ANCOVA-analysed designs.
- Available software computes slope-comparison power but does not optimise the phase allocation.

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@winkens2005 calculated optimal time-point positions for linearly divergent treatment effects under compound symmetry, but did not consider the run-in setting. @yi2002 derived sample sizes for trends across repeated measurements under missing data. @frison1992 showed that having more than one pre-treatment measurement is beneficial for designs analyzed via ANCOVA. @overall1994 provided sample size formulas for repeated measurement designs. @galbraith2002 developed general guidelines for the design of clinical trials with longitudinal outcomes. The slopepower command [@nash2021] and the longpower R package [@iddi2022] implement power calculations for slope-comparison trials but do not optimize the allocation of observations across phases.

2 Problem formulation

2.1 Fixed total visits

[4] **The fixed-visit problem minimises the treatment-effect variance over the phase split under a total-count constraint.**

- The budget J excludes baseline and must be split as $J_0+J_1+J_2 = J$.
- Feasibility requires $J_0 \geq 0$, $J_1 \geq 1$, and $J_2 \geq 0$.
- The objective is the $\gamma\gamma$ entry of $(X'\Sigma^{-1}X)^{-1}$.

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Given J total observations (excluding baseline), choose (J_0^*, J_1^*, J_2^*) to minimize $\mathrm{Var}_1(\hat{\gamma})$ subject to $J_0 + J_1 + J_2 = J$, $J_0 \geq 0$, $J_1 \geq 1$, $J_2 \geq 0$.

The objective function is $V(J_0, J_1, J_2) = [(X'\Sigma^{-1}X)^{-1}]_{\gamma\gamma}$ computed via the Woodbury identity from Paper 1.

2.2 Fixed total duration

[5] **The fixed-duration problem instead constrains total trial length and may also optimise observation timing.**

- Duration is fixed via $T = (J_0+J_1+J_2)\cdot t$, so the count constraint becomes $J_0 + J_1 + J_2 = T/t$.
- The variance $\mathrm{Var}_1(\hat{\gamma})$ is minimised subject to this constraint.
- A general version permits non-equally-spaced observations and optimises over the observation times.

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Alternatively, fix the trial duration $T = (J_0+J_1+J_2)\cdot t$ and minimize $\mathrm{Var}_1(\hat{\gamma})$ subject to $J_0 + J_1 + J_2 = T/t$, or more generally allow non-equally-spaced observations within each phase and minimize over the observation times as well.

3 Diminishing returns from run-in observations

3.1 Marginal value of the J_0-th run-in observation

Define the marginal variance reduction from adding the J_0-th run-in observation:

$$\Delta V_{J_0} = V(J_0 - 1, J_1, J_2) - V(J_0, J_1, J_2) \tag{1}$$

3.2 The $J_0 = 1$ anomaly

[6] **The first run-in observation is less valuable than the second because it introduces a new model parameter.**

- For moderate σ_b/σ, $\Delta V_1 < \Delta V_2$.

Table 1: Marginal variance reduction from the J_0-th run-in observation ($J_1 = 2$, $J_2 = 0$, $\sigma^2 = 1$, $\sigma_b^2 = 1$).

J_0	ΔV_{J_0}
1	0.1765
2	0.6667
3	0.5124
4	0.3176
5	0.2008
6	0.1342
7	0.0945
8	0.0696

- Adding one run-in observation expands the model from two to three parameters, consuming a degree of freedom.
- The second run-in observation adds no parameter and delivers pure information gain.

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An important non-monotonicity is visible in the table: $\Delta V_1 < \Delta V_2$ for moderate σ_b/σ. The first run-in observation is less valuable than the second. This occurs because adding a single run-in observation forces the model from 2 parameters (β, γ) to 3 parameters (δ, β, γ), consuming a degree of freedom. The second run-in observation does not increase the parameter count and provides pure information gain.

[7] **The single-run-in penalty parallels a known ANCOVA result and yields a clear design rule.**

- @frison1992 found a single pre-treatment ANCOVA covariate can reduce power at low pre-post correlation.
- The same degree-of-freedom cost applies in the run-in setting.
- Designers should plan either zero or at least two run-in observations, never exactly one.

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*This effect is analogous to @frison1992's finding that a single pre-treatment measurement used as an ANCOVA covariate can reduce power when the pre-post correlation is low. The practical implication is clear: **a trial designer should plan either zero or at least two run-in observations, not one.***

3.3 Correlation structure governs diminishing returns

The correlation between the J_0-th run-in change score and the treatment-period change scores determines the information gain. Under compound symmetry:

$$\mathrm{Corr}(C_{-J_0}, C_l) = \frac{\sigma_b^2 t^2}{2\sigma^2 + \sigma_b^2 t^2} \tag{2}$$

Table 2: Optimal allocation (J_0^*, J_1^*, J_2^*) for fixed total observations J, by signal-to-noise ratio.

J	Low ($\sigma_b/\sigma = 0.32$)	Med ($\sigma_b/\sigma = 1.0$)	High ($\sigma_b/\sigma = 2.24$)	AD ($\sigma_b/\sigma = 1.70$)
3	(0,3,0)	(0,2,1)	(0,2,1)	(0,2,1)
4	(0,4,0)	(0,2,2)	(0,2,2)	(0,2,2)
5	(0,5,0)	(0,3,2)	(0,3,2)	(0,3,2)
6	(0,6,0)	(0,3,3)	(0,3,3)	(0,3,3)
7	(0,5,2)	(0,4,3)	(0,4,3)	(0,4,3)
8	(0,5,3)	(0,4,4)	(0,4,4)	(0,4,4)

[8] **Under compound symmetry diminishing returns stem from the information matrix, whereas under AR(1) declining correlation accelerates them.**

- The run-in change-score correlation is constant in J_0 under compound symmetry.
- Diminishing returns for $J_0 \geq 2$ therefore arise from the information matrix structure.
- Under AR(1) the correlation falls with temporal distance, yielding faster diminishing returns from distant run-in observations.

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which is constant in J_0. The diminishing returns (for $J_0 \geq 2$) arise from the information matrix structure, not from declining correlation. Under AR(1) [@winkens2007], the correlation does decline with temporal distance, producing faster diminishing returns from distant run-in observations.

4 Optimal allocation for fixed J

5 When is the common close phase beneficial?

[9] **The common close phase aids estimation only indirectly, through the random slope, so its value is conditional.**

- Common close observations inform the random slope b_i.
- They do not contribute to the treatment comparison directly.
- They are beneficial only when the added slope information sufficiently lowers $\mathrm{Var}(\hat{\gamma})$.

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The common close phase contributes information about the random slope b_i without contributing to the treatment comparison directly. It is beneficial when the additional slope information sufficiently reduces the variance of $\hat{\gamma}$.

```
## Minimum sigma_b/sigma for common close benefit:
```

```
##    k=1: sigma_b/sigma >= 0.1
##    k=2: sigma_b/sigma >= 0.1
##    k=3: sigma_b/sigma >= 0.1
##    k=4: sigma_b/sigma >= 0.1
```

5.1 Threshold as a function of J_1

6 Non-equally-spaced designs

6.1 General time points

From the notebook derivations, the variance for the conventional design with general time points t_1, t_2 (under normalization $\sum t_i^2 = 1$, $\sum t_i = 0$) is:

$$\mathrm{Var}(\hat{\gamma}) = \frac{3\sigma^2}{1 - t_1 t_2} + 2\sigma_b^2 \tag{3}$$

[10] **Maximally separated time points minimise the slope-estimation variance, reproducing the classical extremes result.**

- The variance is minimised when $t_1 t_2$ is most negative.
- This occurs when the two time points are maximally separated.
- It recovers the known optimality of observations at the extremes of follow-up for slope estimation.

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This is minimized when $t_1 t_2$ is minimized (most negative), i.e., when the two time points are maximally separated. This reproduces the known result that observations at the extremes of the follow-up period are optimal for slope estimation.

6.2 Optimization over observation times within phases

7 Replicated endpoint observations

7.1 Motivation

[11] **The averaging logic that motivates run-in can be applied symmetrically at the end of treatment to reduce endpoint measurement error.**

- Run-in averages pre-randomisation observations to cut the measurement-error contribution to the baseline slope.
- Replicated measures raise power with diminishing returns past 20-50 replications.
- Closely spaced replicates of the final visit average out endpoint measurement error under the same treatment condition.

To be completed by rgt.

The run-in mechanism averages multiple pre-randomization observations to reduce the contribution of measurement error σ^2 to the estimated baseline slope [@frost2008; @frison1992]. @goulet2019 showed that replicated measures increase power but with diminishing returns past 20-50 replications. The same logic applies at the end of the treatment period: if the final scheduled visit is replaced by (or augmented with) two or three closely-spaced measurements, all under the same treatment condition, the measurement error on the endpoint assessment is reduced by averaging.

[12] **Replicating an endpoint or baseline visit scales down its residual variance while leaving the random-slope contribution intact.**

- Replicating the last visit n_{rep} times over a short interval gives effective residual variance σ^2/n_{rep}.
- The random slope contribution $\sigma_b^2 t^2$ is essentially unchanged.
- Replicating the baseline visit similarly reduces σ^2 in the change-score covariance.

To be completed by rgt.

Concretely, if the last visit at time $J_1 t$ is replicated n_{rep} times over a short interval (days to weeks, negligible relative to t), the effective residual variance for the averaged endpoint becomes σ^2/n_{rep} while the random slope contribution $\sigma_b^2 t^2$ is essentially unchanged. Similarly, the baseline visit at $j = 0$ can be replicated to reduce σ^2 in the change-score covariance.

[13] **Replicated endpoints differ fundamentally from common close observations in timing and the information they provide.**

- Common close observations occur at new time points after treatment stops.
- They inform the off-treatment slope.
- Replicated endpoints sit at the same time point as the last treatment visit and only reduce measurement error.

To be completed by rgt.

This is distinct from the common close design (Section 4): common close observations are at new *time points after treatment stops, providing information about the off-treatment slope. Replicated endpoints are at the* same *time point as the last treatment visit, providing only measurement-error reduction.*

7.2 Variance formula with replication

[14] **Replication enters only through modified entries of R, leaving the Woodbury computation otherwise unchanged.**

- The diagonal entry for a replicated visit uses σ^2/n_{rep} in place of σ^2.

Table 3: Fractional variance reduction from replicating the final endpoint ($J_0 = 0$, $J_1 = 2$, $J_2 = 0$).

n_{rep}	$\sigma_b/\sigma = 0.5$	$\sigma_b/\sigma = 1$	$\sigma_b/\sigma = 2$	$\sigma_b/\sigma = 5$
1	0.000	0.000	0.000	0.000
2	0.182	0.091	0.030	0.005
3	0.250	0.125	0.042	0.007
5	0.308	0.154	0.051	0.009

- Off-diagonal entries reflecting shared baseline error use $\sigma^2/n_{\text{rep,pre}}$.
- The remaining Woodbury computation proceeds identically.

To be completed by rgt.

The residual covariance matrix R is modified so that the diagonal element corresponding to a replicated visit uses σ^2/n_{rep} instead of σ^2, and the off-diagonal elements reflecting the shared baseline error use $\sigma^2/n_{rep,pre}$. The Woodbury computation proceeds identically.

[15] **Endpoint replication helps most when measurement error dominates and little when slope variability does.**

- The benefit is largest at small σ_b/σ.
- At large $\sigma_b/\sigma > 2$ the gain is modest.
- Variance is then dominated by $\sigma_b^2 t^2$, which replication cannot reduce.

To be completed by rgt.

The table confirms that endpoint replication is most beneficial when measurement error dominates (σ_b/σ small). When between-subject slope variability is large ($\sigma_b/\sigma > 2$), the gain from replication is modest because the variance is dominated by $\sigma_b^2 t^2$, which replication cannot reduce.

7.3 Replication vs additional run-in observations

[16] **A fixed budget of extra measurements can be allocated either to run-in observations or to endpoint replicates.**

- Run-in observations sit at distinct time points separated by t.
- Endpoint replicates are closely spaced at one time point.
- Replicates provide measurement-error reduction only, not slope information.

To be completed by rgt.

A natural question is whether a fixed budget of additional measurements should be spent on run-in observations (at distinct time points, separated by t) or on endpoint replicates (closely spaced, measurement-error reduction only).

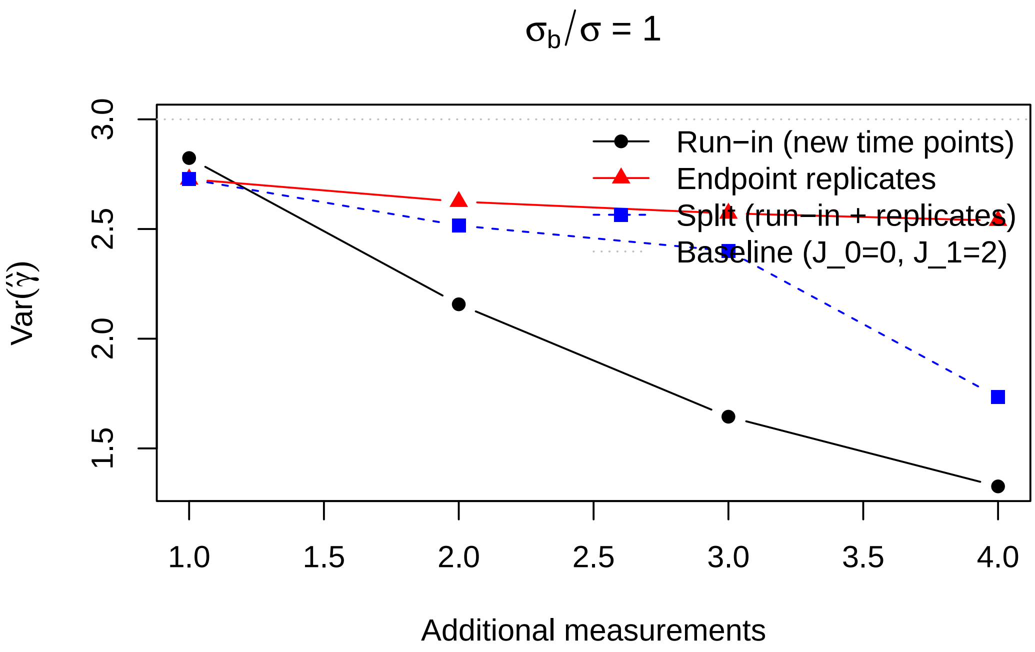

Figure 1: (#fig:rep_vs_runin)Variance as a function of additional measurements allocated either to run-in (new time points) or endpoint replication (same time point). Baseline: $J_0 = 0$, $J_1 = 2$, $\sigma^2 = 1$, $\sigma_b^2 = 1$.

[17] **Whether run-in or endpoint replication wins depends on the signal-to-noise ratio.**

- At $\sigma_b/\sigma \geq 1$, run-in observations dominate.
- They supply information about the individual slope b_i, not only measurement-error reduction.
- At $\sigma_b/\sigma < 1$, replication is competitive because measurement error dominates the variance.

To be completed by rgt.

When $\sigma_b/\sigma \geq 1$, run-in observations at new time points outperform endpoint replicates because they provide information about the individual slope b_i, not just measurement-error reduction. When $\sigma_b/\sigma < 1$, endpoint replication can be competitive because measurement error is the dominant source of variance.

7.4 Combined replication at both ends

[18] **Replicating both terminal visits yields symmetric measurement-error reduction.**

- Baseline replication reduces σ^2 at the start of follow-up.
- Endpoint replication reduces σ^2 at the end of treatment.
- Together they lower measurement error symmetrically at both ends.

To be completed by rgt.

Table 4: Per-pair variance with baseline and endpoint replication ($J_0 = 0$, $J_1 = 2$, $\sigma^2 = 1$, $\sigma_b^2 = 1$).

n_{pre}	$n_{\text{post}} = 1$	$n_{\text{post}} = 2$	$n_{\text{post}} = 3$
1	3.0000	2.7273	2.6250
2	2.7273	2.5000	2.4138
3	2.6250	2.4138	2.3333

Replicating both the baseline and endpoint visits provides symmetric measurement-error reduction.

8 Incorporating dropout

8.1 Pattern-mixture approach

[19] **Dropout is accommodated by a pattern-mixture model that combines stratum-specific sample sizes.**

- The approach follows @frost2008 Section 2.3 and related pattern-mixture work.
- Participants are stratified by their observed visit pattern.
- The overall requirement is a weighted average across strata.

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Following @frost2008 (Section 2.3), dropout is handled via a pattern-mixture model [@dawson1993; @lu2008]. Participants are stratified by their observed visit pattern, and the overall variance is a weighted average:

$$N = \frac{1}{\sum_s p_s / N_s} \tag{4}$$

[20] **The pattern-mixture formula is a conservative approximation that exact monotone-dropout treatments improve upon.**

- Here p_s is the proportion with pattern s and N_s the size needed if all had pattern s.
- The approximation treats each stratum independently.
- @hu2021 derive exact formulas under monotone dropout, more rigorous and less conservative when dropout is informative about the random slope.

To be completed by rgt.

where p_s is the proportion with pattern s and N_s is the sample size that would be needed if all participants had pattern s. This approximation treats each stratum independently. @hu2021 derive exact variance formulas under monotone dropout for the three-component random coefficient regression

Table 5: Sample size adjusted for dropout via pattern-mixture ($J_0 = 2$, $J_1 = 2$, $J_2 = 1$, MIRIAD parameters).

Annual dropout	N (no dropout)	N (with dropout)
0.0	104	104
0.1	104	113
0.2	104	121
0.3	104	128

Table 6: Top 10 allocations for $J = 5$ observations (MIRIAD parameters, $\Delta = 0.25$, 90% power).

(J_0, J_1, J_2)	$\mathrm{Var}_1(\hat{\gamma})$	N per group
(0,3,2)	0.291783	50
(0,2,3)	0.303230	51
(1,2,2)	0.453050	77
(2,3,0)	0.551570	93
(0,4,1)	0.560024	95
(1,3,1)	0.587888	99
(3,2,0)	0.593958	100
(2,2,1)	0.616148	104
(0,1,4)	0.701145	118
(1,1,3)	0.803845	136

model ($\sigma^2, \sigma_a^2, \sigma_b^2$), accounting for the effect of dropout on the estimability of the random slope variance. Their approach is more rigorous than the pattern-mixture approximation used here, which is conservative when dropout is informative about the random slope.

9 Numerical examples

9.1 AD neuroimaging trial

[21] **The AD example evaluates optimal allocation under MIRIAD parameters across several budgets and dropout scenarios.**

- Parameters are $\sigma^2 = 0.3363$, $\sigma_b^2 = 0.9745$, and $t = 1$ year.
- Allocation is compared for $J = 4, 5, 6$ total observations.
- Both with-dropout and without-dropout cases are considered.

To be completed by rgt.

Using MIRIAD parameters ($\sigma^2 = 0.3363$, $\sigma_b^2 = 0.9745$, $t = 1$ year), compare the optimal allocation for $J = 4, 5, 6$ total observations with and without dropout.

10 Discussion

11 Conflict of interest

[22] **The author reports no conflicts of interest.**

- No financial conflicts are present.
- No non-financial conflicts are present.
- No competing interests bear on this work.

To be completed by rgt.

The author declares no conflicts of interest.

12 Funding

[23] **The work received no specific funding.**

- No external grant supported the study.
- No internal grant supported the study.
- No sponsor influenced the analysis.

To be completed by rgt.

This work received no specific funding.

13 Data availability statement

[24] **The implementing package, tests, and shared bibliography are openly available in the project repository.**

- Allocation routines reside in allocation.R, replicated.R, and design.R.
- Validation tests and the four-paper bibliography are included.
- Specific paths appear in the Reproducibility section below.

To be completed by rgt.

The ‘runinpower‘ R package implementing the optimal-allocation routines (‘allocation.R‘, ‘replicated.R‘, ‘design.R‘), validation tests, and the shared bibliography across the four-paper compendium are openly available in the project repository. Specific paths are listed in the Reproducibility section below.

14 Reproducibility

[25] **The manuscript build relies on a small, fixed set of R packages.**

- Rendering used R 4.4.x.

- The in-package runinpower source and tinytest provide computation and testing.
- rmarkdown and bookdown drive the manuscript build.

To be completed by rgt.

This manuscript was rendered with R 4.4.x. Principal packages used are the in-package ‘runinpower‘ source, ‘tinytest‘ (testing harness), and ‘rmarkdown‘ plus ‘bookdown‘ for the manuscript build.

[26] **Results are deterministic, with seeding applied only where Monte Carlo benchmarking is used.**

- Optimisation outputs are deterministic given fixed input parameters.
- Monte Carlo benchmarking, where reported, is reproducible.
- The driver sets set.seed(20260418) at its entry point.

To be completed by rgt.

***Random-number generator.** Optimisation results are deterministic given fixed input parameters. Where Monte Carlo benchmarking is reported, the driver sets ‘set.seed(20260418)‘ at its entry point.*

[27] **Source code, companion manuscripts, and the shared bibliography are located at documented repository paths.**

- The R package source is under R/.
- Companion manuscripts (Papers 1, 2, 4) are under analysis/report/{01,02,04}-*/.
- This manuscript source and the shared bibliography are under analysis/report/03-allocation/.

To be completed by rgt.

***Data and code availability.** The R package source is at ‘R/‘. The companion manuscripts (Papers 1, 2, 4) are at ‘analysis/report/01,02,04-*/‘. The manuscript source and shared bibliography (‘references.bib‘, a symlink to ‘../../references.bib‘) are at ‘analysis/report/03-allocation/‘.*

15 References

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